# Replication of Published Longevity-Associated Variants in the French Centenarians Cohort

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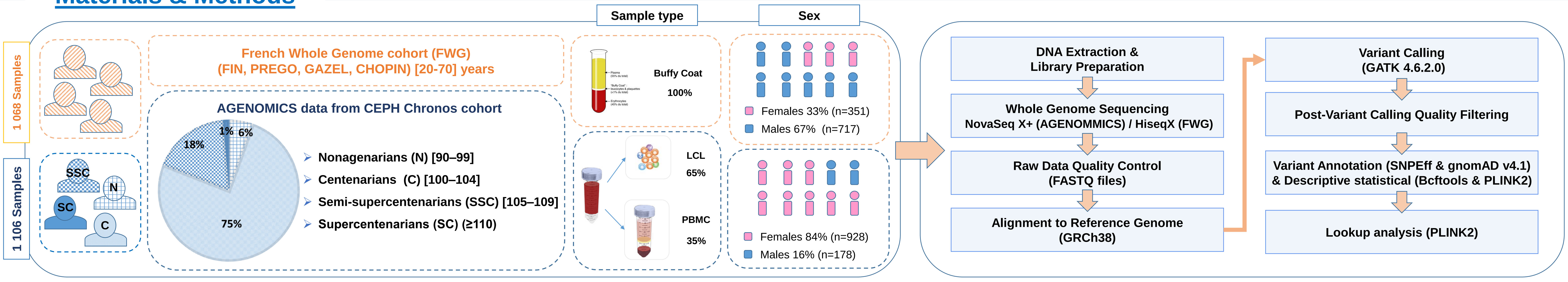
## Introduction

Aging is a universal and complex process driven by both genetic and environmental factors, many of which contribute to age-related diseases, making centenarians a valuable model of healthy aging and disease resistance [1]. Numerous genetic variants have been associated with exceptional longevity across various studies, cohorts, and platforms, yet their replication in independent populations remains limited. The AGENOMICS project aims to uncover the genetic determinants of exceptional longevity through whole-genome sequencing of over **1 100 French centenarians** from the CEPH Chronos cohort [2]. In this context, **we performed a targeted lookup of literature-reported variants associated with longevity** [3] or Alzheimer's disease (AD) [4], and compared their allele frequencies with those from a FranceGenRef (FWG) control cohort [5], to assess the replicability of these associations in a French population.

## Materials & Methods




AGENOMICS samples, stratified into four age groups, include 35% peripheral blood mononuclear cells (PBMCs) and 65% lymphoblastoid cell lines (LCLs), with 84% female participants. DNA was sequenced on the NovaSeq X+ platform and processed through alignment, variant calling, and quality control filtering (VQSR, gnomAD filters, complex genomic regions), retaining approximately 30 million variants that passed QC filters. Variants were annotated using SNPEff and gnomAD v4.1, and statistical analyses were performed with Bcftools 1.18 and PLINK2 [6]. These 19 literature-reported variants associated with longevity or neurodegenerative diseases was conducted by querying their allele frequencies in the dataset and comparing them against **1 068 samples** from the FWG cohort (FIN, PREGO, GAZEL, CHOPIN), using a generalized logistic model (GLM) with two covariate models: PCs combined with sex (Model 1), principal components (PCs) only (Model 2).

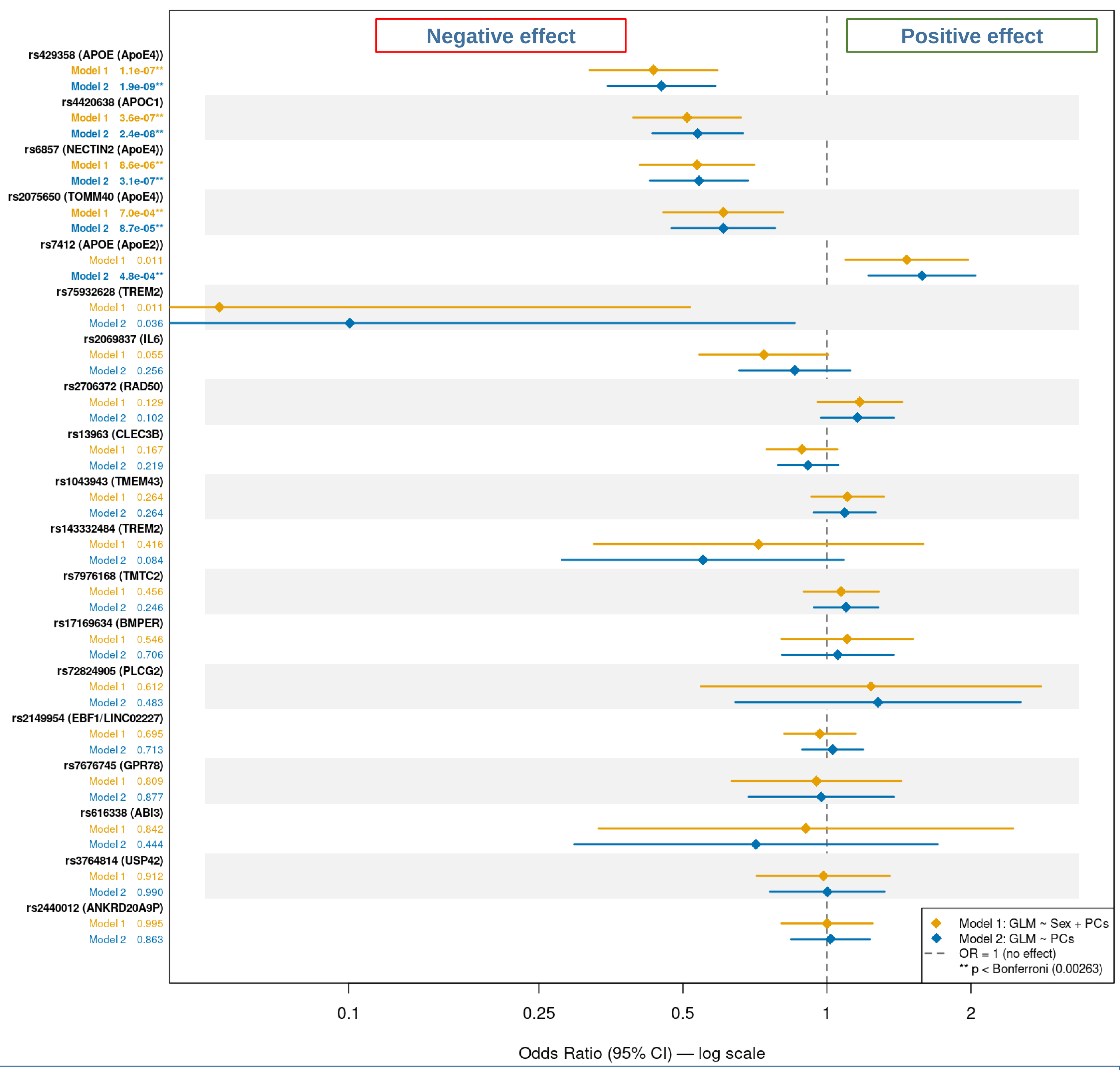
## Results


AGENOMICS and FWG cohorts partially overlap on PC1–PC2, reflecting minor population structure and sequencing platform differences, while full overlap on PC3–PC4 confirms overall genetic similarity. European ancestry was confirmed by projection onto 1000 Genomes populations.

**Table legend:** Variant ID: SNP identifier; Chr: chromosome; AF (ALT): alternative allele frequency

| Variant ID | Chr | Implicated gene | Ancestry (Study) | Genotyping method | P-value (Study) | AF (ALT) AGENOMICS | AF (ALT) FWG | AF (ALT) gnomAD |
|---|---|---|---|---|---|---|---|---|
| rs13963 | 3 | CLEC3B | East Asian (Japanese) | Genotyping | $1.87 \times 10^{-8}$ | 0.39295 | 0.41776 | 0.43411 |
| rs1043943 | 3 | TMEM43 | East Asian (Han Chinese) | Genotyping | $3.59 \times 10^{-8}$ | 0.58762 | 0.61075 | 0.65151 |
| rs7676745 | 4 | GPR78 | European | Imputation | $4.30 \times 10^{-8}$ | 0.04652 | 0.04486 | 0.05166 |
| rs2149954 | 5 | EBF1/LINC02227 | European | Imputation | $1.74 \times 10^{-8}$ | 0.39792 | 0.40327 | 0.36291 |
| rs2706372 | 5 | RAD50 | European | Genotyping | $5.42 \times 10^{-8}$ | 0.22042 | 0.19112 | 0.32581 |
| rs143332484 | 6 | TREM2 | Multiple cohorts | Mixed | $1.55 \times 10^{-14}$ | 0.00903 | 0.01355 | 0.00708 |
| rs75932628 | 6 | TREM2 | Multiple cohorts | Mixed | $5.38 \times 10^{-24}$ | 0.00090 | 0.00608 | 0.00199 |
| rs17169634 | 7 | BMPER | East Asian (Han Chinese) | Mixed | $1.45 \times 10^{-10}$ | 0.08943 | 0.07243 | 0.17058 |
| rs2069837 | 7 | IL6 | East Asian (Han Chinese) | Genotyping | $4.05 \times 10^{-8}$ | 0.08266 | 0.07944 | 0.09631 |
| rs3764814 | 7 | USP42 | European | Imputation | $5 \times 10^{-15}$ | 0.07633 | 0.08131 | 0.19174 |
| rs7976168 | 12 | TMTC2 | European | Imputation | $4 \times 10^{-9}$ | 0.33379 | 0.32290 | 0.31255 |
| rs2440012 | 13 | ANKRD20A9P | East Asian (Han Chinese) | Genotyping | $4.89 \times 10^{-8}$ | 0.18202 | 0.18879 | 0.00143 |
| rs72824905 | 16 | PLCG2 | Multiple cohorts | Mixed | $5.38 \times 10^{-10}$ | 0.01265 | 0.00794 | 0.00494 |
| rs616338 | 17 | ABI3 | Multiple cohorts | Mixed | $4.56 \times 10^{-10}$ | 0.99503 | 0.99019 | 0.99423 |
| rs4420638 | 19 | APOC1 | European | Genotyping | $2.2 \times 10^{-16}$ | 0.09214 | 0.15841 | 0.18007 |
| rs7412 | 19 | APOE (APOE2) | European | Mixed | $1.7 \times 10^{-12}$ | 0.10614 | 0.07850 | 0.07784 |
| rs429358 | 19 | APOE (APOE4) | European | Mixed | $1.3 \times 10^{-36}$ | 0.05601 | 0.12336 | 0.15736 |
| rs6857 | 19 | NECTIN2 (APOE4) | European | Imputation | $2 \times 10^{-27}$ | 0.07453 | 0.13645 | 0.12608 |
| rs2075650 | 19 | TOMM40 (APOE4) | European/ East Asian (Han Chinese) | Genotyping | $3.39 \times 10^{-17}$ | 0.06594 | 0.11589 | 0.13214 |

- The table below presents **19 variants associated with longevity or AD in the literature**, identified by genotyping or imputation in European and East Asian cohorts (p-values: $1.3\times10^{-36}$ to $5.42\times10^{-8}$). Variants with a positive effect on lifespan are shown in green, variants with a negative effect in red; highlighted rows failed quality control and should be interpreted with caution.
- The forest plot shows odds ratios (95% CI) under two models: Model 1 (GLM ~ Sex + PCs) and Model 2 (GLM ~ PCs only). **Five variants reached Bonferroni significance** ($p < 0.00263$): rs429358 (**APOE4**), rs4420638 (**APOC1**), rs6857 (**NECTIN2**) and rs2075650 (**TOMM40**) showed OR < 1 (negative effect), while rs7412 (**APOE2**) showed OR > 1 (positive effect).
- The four variants with a negative effect replicated under both models; APOE2 replicated under Model 2 only, likely due to sex imbalance between the AGENOMICS cohort and controls. **Of these five, only rs4420638, rs7412 and rs2075650 passed QC filters**; rs6857 failed due to low mappability in a segmental duplication region, and rs429358 failed the Quality by depth (QD) filter.
- **The remaining 14 variants did not replicate**, potentially due to platform differences, insufficient power, or population specificity. Notably, ANKRD20A9P, BMPER, IL6, CLEC3B and TMEM43 were originally identified in East Asian cohorts and may harbor population-specific effects.
- **TREM2, PLCG2 and ABI3** were not replicated for longevity per se, but their OR in our cohort are consistent with their reported AD risk effect in the literature.




## Conclusion and Future Perspectives

This study demonstrates the potential of the AGENOMICS cohort for longevity research. The successful replication of three variants, **rs4420638 (APOC1), rs7412 (APOE2)** and **rs2075650 (TOMM40/APOE4),** confirms their robust association with longevity across independent cohorts. The limited replication of the remaining variants may reflect **differences in genotyping platforms versus WGS, statistical approaches, and population ancestry**, as several variants were originally identified in East Asian cohorts and may not generalize to the French population studied here. Future analyses should prioritize a genome-wide association study (GWAS) across all variants available in the AGENOMICS cohort, which could reveal novel longevity-associated loci and contribute to a broader understanding of the genetics of human longevity. Additionally, sex-stratified replication analyses could help clarify the observed sex-dependent effects on variant associations.

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